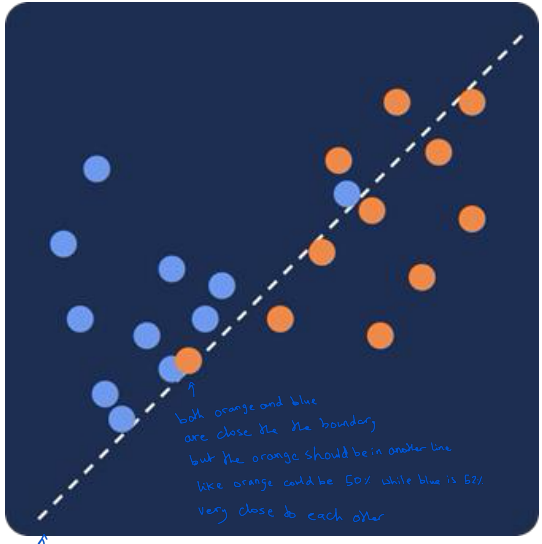
Classification

finite number of catagories

ex: pics of animals like classifying a cat,
yes/no, true/false, low/high risks

depending on the features, classification is supervisor learning because the output is known



Instructor: Dr. Heyfa Ammar

PURPOSE

What we will learn today

The framework

How a classifier turns features into a class label

Zero-one loss

Counting mistakes, and the classifier that makes the fewest

Bayes error rate

The error that no classifier can avoid

Metrics

Judging a classifier with more than accuracy

Along the way: two quick polls and one practice problem.

Classification Methods

These are supervised learning methods in which a categorical response variable Y takes one of c possible values (for example whether a person is sick or healthy), which is to be predicted from a vector $\boldsymbol{X}$ of explanatory variables (for example, the blood pressure, age, and smoking status of the person), using a prediction function g.

or features (x)

In this sense, g classifies the input $\boldsymbol{X}$ into one of the classes, say in the set $\{0, \ldots, c-1\}$. For this reason, we will call g a classification function or simply classifier.

As with any supervised learning technique, the goal is to minimize the expected loss or risk

$$\ell(g) = \mathbb{E}\,\mathrm{Loss}(Y, g(\boldsymbol{X})) \tag{1}$$

difference between true y, and predicted

for some loss function, $\mathrm{Loss}(y, \widehat{y})$, that quantifies the impact of classifying a response y via $\widehat{y} = g(\boldsymbol{x})$.

A function we can use for classification problem

Zero-one Loss

No difference between true and predicted will be 0 like a model choosing a pic of an animal is a cat
If there is a difference then its 1

The natural loss function is the zero–one (also written 0–1) or indicator loss: $\text{Loss}(y, \widehat{y}) := \mathbb{I}\{y \neq \widehat{y}\}$; that is, there is no loss for a correct classification ($y = \widehat{y}$) and a unit loss for a misclassification ($y \neq \widehat{y}$).

Theorem: Optimal classifier

For the loss function $\text{Loss}(y, \widehat{y}) = \mathbb{I}\{y \neq \widehat{y}\}$, an optimal classification function is

$$g^*(\boldsymbol{x}) = \underset{y \in \{0,\ldots,c-1\}}{\text{argmax}} \; \mathbb{P}[Y = y \mid \boldsymbol{X} = \boldsymbol{x}]. \quad (2)$$

$g^*(\boldsymbol{x})$: best model

Given a specific set of features, what is the output like it correspond to a cat not dog like if its more probability its a cat like 90% sure, it is a cat

it doesnt output how sure it is, it just output what it thinks

THINK, PAIR, SHARE · 1 MINUTE

Two fingerprint scanners, both about 99.5% accurate

SCANNER A

99.55%

accurate

misclassifies it 0.45%.

SCANNER B

99.54%

accurate

misclassifies 0.46%.

Which scanner would you trust to keep a secret military base safe?

A because its accuracy is a bit higher

However accuracy alone isn't the best to choose just from it, we need to check other features and B could be better

THE FRAMEWORK

From features to a class label

Features X
Blood pressure, age, smoking status

→

Classifier g
A prediction function fitted to data

→

Class $\hat{y} = g(x)$
One of {0, …, c − 1}, such as healthy or sick

The response Y is categorical: it takes one of c possible values: healthy or sick ?

2 classes

This is supervised learning, so g is learned from data where the true class is known.

THE FRAMEWORK

Success is measured by expected loss

$$\ell(g) = \mathrm{E}\,\mathrm{Loss}(Y, g(X))$$

The risk of g: the expected loss

Loss(y, ŷ)
The price of predicting ŷ when the truth is y

Risk ℓ(g)
The average price over all cases g may meet

The goal
Find the classifier g with the smallest risk

Other loss functions do regression, want use MAE, MSE, RMSE

ZERO-ONE LOSS

The natural loss: right or wrong

$$\text{Loss}(y, \hat{y}) = I\{y \neq \hat{y}\}$$

Correct: $y = \hat{y}$
Loss 0. No penalty.

Mistake: $y \neq \hat{y}$
Loss 1. One unit of penalty.

So the risk becomes the probability of misclassifying a new case.

THEOREM · OPTIMAL CLASSIFIER

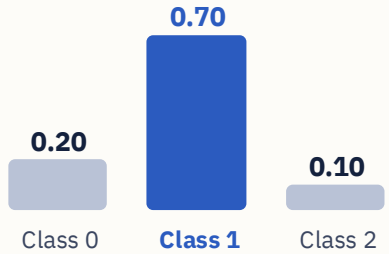
Pick the most probable class

$$g^*(x) = \operatorname{argmax} P[Y = y \mid X = x]$$

taken over $y \in \{0, \ldots, c - 1\}$

For zero-one loss, the best classifier gives x the class with the highest conditional probability.

Illustration: probabilities for one input x



Class 1 has the highest bar, so $g^*(x) = 1$

PROOF IDEA

Why the argmax is optimal

1

Condition on X

$\ell(g) = \mathbb{E}(\, P[Y \neq g(X) \mid X] \,)$

This is the tower property.

2

Fix one x

Minimizing the risk means maximizing P[Y = g(x) | X = x] for every fixed x.

3

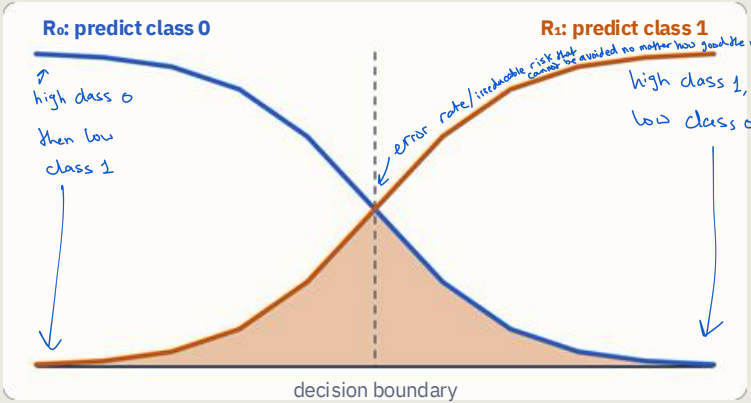
Choose the winner

Let g(x) be the label y whose probability P[Y = y | X = x] is largest.

■

BAYES ERROR RATE

The limit of any classifier



The rule splits the feature space into c regions, one per class.

Where the classes overlap, even the best rule sometimes errs (shaded).

Bayes error rate

$$\ell(g^*) = P[Y \neq g^*(X)]$$

The irreducible risk.

binary classification (2 classes)

The sum of probability should be 1, If the model is 70% its a cat then 30% its a dog

Bayes error Loss

The formulation (2) allows for "ties", when there is an equal probability between optimal classes for a feature vector $\boldsymbol{x}$.

The optimal prediction function depends on the conditional pdf $f(y \mid \boldsymbol{x}) = \mathbb{P}[Y = y \mid \boldsymbol{X} = \boldsymbol{x}]$.

Since we assign $\boldsymbol{x}$ to class y if $f(y \mid \boldsymbol{x}) \geqslant f(z \mid \boldsymbol{x})$ for all z, we do not need to learn the entire surface of the function $f(y \mid \boldsymbol{x})$.

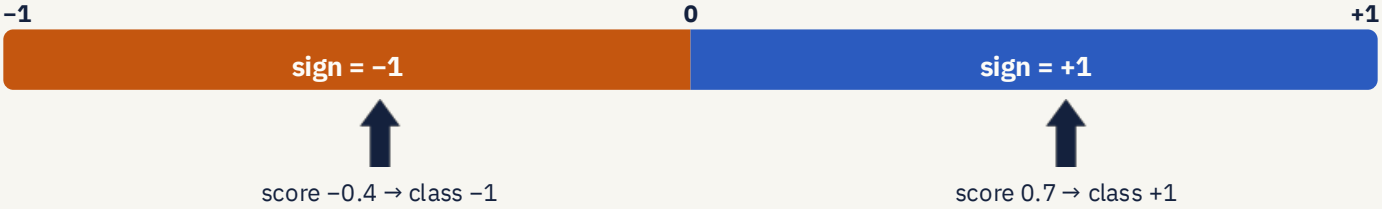
In fact, the assignment (2) divides the feature space into c regions, $\mathcal{R}_y = \{\boldsymbol{x} : f(y \mid \boldsymbol{x}) = \max_z f(z \mid \boldsymbol{x})\}$, $y = 0, \ldots, c-1$.

For the indicator loss, the smallest possible expected loss (i.e., the irreducible risk $\ell(g^*)$) is equal to $\mathbb{P}[Y \neq g^*(\boldsymbol{X})]$. This is called the Bayes error rate.

PRE-CLASSIFIER

A score first, then a label

A pre-classifier can return any real number. For labels −1 and 1 it returns a score in [−1, 1] and the classifier is sign(g).



1) calculate the score
2) see which class it fits

ROADMAP

Ways to fit a classifier

01
Bayesian classification

02
Linear and quadratic discriminant analysis

03
Logistic classification

04
K-nearest neighbors

05
Support vector machines

06
Classification trees

07
Neural networks

Today: how to judge every one of them

TRAINING AND TEST SETS

judge how accurate the model is with specific data

Fit on one set, judge on another

Training set	Test set
τ	τ'

Fit: minimize the training loss

$$\ell\tau(g) = (1/n)\ \Sigma\ \mathrm{Loss}(y(i), g(x(i)))$$

Sum over the n training pairs, over a class of functions G.

Judge: estimate the risk on new data

The training loss is often a poor estimator of the risk.

So we compute the same average on an independent test set τ'.

Two matrices, one loss

Loss matrix L (zero-one loss)

Give a cost to any misclassifications as 1
correctly classified would be 0

actual ↓	0	1	2
0	0	1	1
1	1	0	1
2	1	1	0

(100% penalty of classifying 1 as 0)

Columns: predicted class

3 class (0, 1, 2)
how many time we classified 0 as 0, 1 as 1, 2 as 2

Confusion matrix M (counts)

Counts the number of detections (correct or incorrect)

actual ↓	0	1	2
0	M(0,0)	M(0,1)	M(0,2)
1	M(1,0)	M(1,1)	M(1,2)
2	M(2,0)	M(2,1)	M(2,2)

M(j,k): times actual class j was predicted as k

$$\text{loss} = (1/n)\,\Sigma\,[L \odot M](j,k) = 1 - \text{tr}(M)/n \text{ for zero-one loss}$$

CONFUSION MATRIX

Three classes: dog, cat, possum

Actual ↓	PREDICTED: Dog	Cat	Possum
Dog	30	2	6
Cat	8	22	15
Possum	7	4	41

30: correctly classified 30 image of dog

2: 2 misclassified dog as cat

6: 6 misclassification dog as possum

Reading the matrix

Rows are actual classes, columns are predicted classes. Blue cells on the diagonal are correct.

Correct: 30 + 22 + 41 = 93 of n = 135

Error = 1 − 93/135 ≈ 31.1%

or adding each error / total

Your turn: which single mistake is the most common?

Clinical classifier, when penalty>1

Worked example with the clinic classifier (false alarm costs 1, missed sick patient costs 10):

Cell	Count	Cost each	Contribution
tn (healthy → healthy)	1033	0	0
tp (sick → sick)	240	0	0
fp (healthy → sick)	91	1	91
fn (sick → healthy)	136	10	1360

Correctly classified ⇒ no cost

incorrect classification ⇒ has cost

1 x 91

10 x 10

Loss = (91 + 1360) / 1500 = **0.967**

Notice that the 136 missed patients are only 9% of the mistakes' count, but they make up 94% of the cost. With zero-one loss the same classifier scores (91 + 136) / 1500 = 0.151, which is just the error rate.

zero-one loss: $Loss_{0-1} = \frac{\sum \text{Misclassification}}{\text{Total}}$

$Loss_{0-1} = \frac{91 \times 1 + 136 \times 1}{1500} = 0.151$

$Loss = \frac{91 \times 1 + 136 \times 10}{1500} = 0.967$

(more penalty on the critical cost)

Positive class & class we want to predict

TRUE AND FALSE, POSITIVE AND NEGATIVE

Four counts for the Dog class

Actual ↓	PREDICTED: Dog	Cat	Possum
Dog	30 TP	2 FN	6 FN
Cat	8 FP	22 TN	15 TN
Possum	7 FP	4 TN	41 TN

True positive (TP) = 30 Correct prediction the positive class
Dogs correctly called dogs

False negative (FN) = 2 + 6 = 8 Falsely classifying the incorrect class
(False a/rong class, should be positive)
Dogs called cat or possum (rest of the row)

Row

False positive (FP) = 8 + 7 = 15 classifying as dog → positive
Cats or possums called dogs (rest of the column)

Col

True negative (TN) = 22 + 15 + 4 + 41 = 82
Everything outside the Dog row and column

Classified other as a non dog correctly

Classification is supervised bcuz we have the label (output)

Last slide for classified

COUNTS FOR CLASS J

From the matrix to four formulas

True positive

$$tp(j) = M(j,j)$$

The diagonal entry

False positive

$$fp(j) = \Sigma\, M(k,j),\ k \neq j$$

Column sum without the diagonal

False negative

$$fn(j) = \Sigma\, M(j,k),\ k \neq j$$

Row sum without the diagonal

True negative

$$tn(j) = n - fn(j) - fp(j) - tp(j)$$

Everything that is left

Memory aid: false positives live in the column, false negatives live in the row.

BINARY CLASSIFICATION

Misclassification error and accuracy

	PREDICTED	
Actual ↓	class j	other
class j	**tp** correct	**fn** mistake
other	**fp** mistake	**tn** correct

Error: the orange cells

$$\text{error}(j) = (\text{fp}(j) + \text{fn}(j)) \,/\, n$$

(false positives + false negatives) / n

Accuracy: the blue cells

$$\text{accuracy}(j) = (\text{tp}(j) + \text{tn}(j)) \,/\, n$$

accuracy = 1 − error. With two classes, fp(0) + fn(0) = fp(1) + fn(1), so it does not matter which class you call j.

PRACTICE · 2 MINUTES IN PAIRS

Your turn: the Cat class

Treat Cat as the positive class. Find:

tp fp fn tn

Check yourself: the four numbers must add up to n = 135.

$T_p = 22$

$F_p = 2+4 = 6$

$F_n = 8+15 = 23$

$T_n = 30+6+7+41 = 84$

ANSWER

The four counts for Cat

Actual ↓	PREDICTED Dog	Cat	Possum
Dog	30 TN	2 FP	6 TN
Cat	8 FN	22 TP	15 FN
Possum	7 TN	4 FP	41 TN

tp = 22
Cats called cats

fp = 2 + 4 = 6
Dogs and possums called cats

fn = 8 + 15 = 23
Cats called dogs or possums

tn = 30 + 6 + 7 + 41 = 84
Everything outside the Cat row and column

Check: 22 + 6 + 23 + 84 = 135 = n

OTHER CLASSIFICATION METRICS

When accuracy is not enough

Top-secret facility

A fingerprint system decides who is authorized. A false positive lets an intruder in, which is extremely dangerous.

Online retail discount

A fingerprint system decides who gets a discount. A false positive just makes a customer happy.

Both are binary problems, but a false positive costs very different amounts.

OTHER CLASSIFICATION METRICS

A 99.55% accurate guard

Actual ↓	PREDICTED: Authorized	PREDICTED: Not authorized
Authorized	100 TP	400 FN
Not authorized	50 FP	100,000 TN

99.55

accuracy

number of the correctly classified things over total
so (true positive + true negative)/total

(100 + 100,000) / (100 + 100,000 + 50 + 400) ≈ 99.55%

⚠ **Yet 50 non-authorized people were let in.**

FIX 1

Make the costly mistake cost more

Indicator loss: every mistake costs 1

actual ↓	Authorized	Not auth.
Authorized	0	1
Not auth.	1	0

Locking out a guest and admitting an intruder are equally bad.

Custom loss: an intruder costs 1000

actual ↓	Authorized	Not auth.
Authorized	0	1
Not auth.	1000	0

less critical so kept as 1

more penalty to critical decision

Predicting authorized for a non-authorized person now costs 1000.

Another route: keep the indicator loss and look at more metrics, coming next.

FIX 2 · ADDITIONAL METRICS

Four questions to ask about class j

Precision (positive predictive value)

$tp / (tp + fp)$

Of everything called j, how much really is j?

Ability to decide/classify correctly. Am I right classifying correctly?

Recall (sensitivity)

$tp / (tp + fn)$

out the num of positive interpretation

Of all real j-objects, how many did we find?

Specificity focus on the negative class

$tn / (fp + tn)$

Of all non-j objects, how many did we correctly reject?

Fβ score

$(\beta^2 + 1)tp / ((\beta^2 + 1)tp + \beta^2 fn + fp)$

One number that combines precision and recall.

$\text{Accuracy} = \frac{T_p + T_n}{N}$ $\frac{\text{True pos + True neg}}{\text{Total}}$

POLL · COMPARING CLASSIFIERS

Which one guards the facility?

Classifier 1

Positive (1st col) ↓

actual ↓	Authorized	Not auth.
Authorized	100	400
Not auth.	50	100,000

$Acc1 = \frac{100+100000}{100+100000+400+50} = 99.55\%$

Classifier 2

actual ↓	Authorized	Not auth.
Authorized	50	10
Not auth.	450	100,040

$Acc2 = \frac{50+100040}{50+100040+450+10} = 99.54\%$

Vote: Classifier 1 or Classifier 2? Convince a neighbour in one sentence.

$\text{Precision}1 = \frac{100}{100+50} = 66.7\% = 0.667$

$\text{Precision}2 = \frac{50}{50+450} = 10\%$

COMPARING CLASSIFIERS

Same accuracy, different behavior

Metric	Classifier 1	Classifier 2
Accuracy	0.9955	0.9954
Precision	0.6667	0.1000
Recall	0.2000	0.8333
Specificity	0.9995	0.9955
F1 score	0.3077	0.1786

(higher recall so finds the TP & FN better than the 1st classifier)

Accuracy barely separates them. Precision and recall tell very different stories.

VERDICT

We prefer Classifier 1

CLASSIFIER 1 · PREFERRED

0.667

precision: 2 in 3 accepted people are authorized

Lets in 50 intruders. Finds 20% of authorized people.

CLASSIFIER 2

0.100

precision: 1 in 10 accepted people is authorized

Lets in 450 intruders. Finds 83% of authorized people.

Here a false positive is the costly mistake, so high precision wins. With a different cost of mistakes, the choice could change.

MULTILABEL CLASSIFICATION

When labels can go together

Standard: exactly one class

The classes exclude each other. A satellite image is one of:

clear **cloudy** **foggy**

Multilabel: any subset of labels

The response is a subset Y of {0, ..., c − 1}. An image can also contain:

road **river**

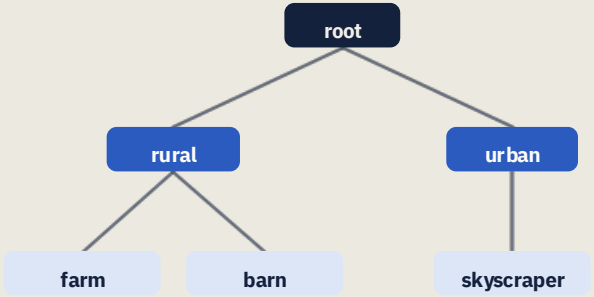
The same response as a binary vector of length c = 5

clear	cloudy	foggy	road	river
1	0	0	1	1

HIERARCHICAL CLASSIFICATION

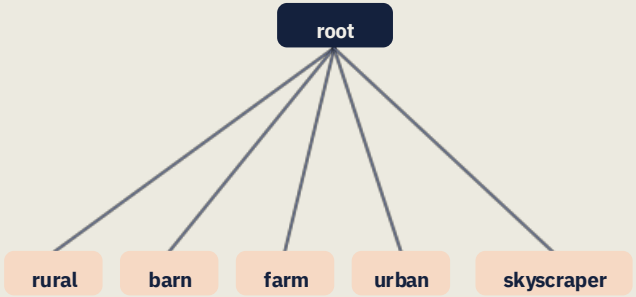
Labels that belong together

Hierarchical



The tree records that farms and barns are rural, skyscrapers are urban.

Non-hierarchical



Five separate labels: the connection between them is missing.

MULTILABEL CLASSIFICATION

Metrics for sets of labels

The prediction $\hat{Y}$ and the true response Y are both subsets of {0, ..., c – 1}. Y(i) is the true set for object i.

Exact match ratio

$\Sigma\, I\{\hat{Y}(i) = Y(i)\} / n$

Strict: the whole set must match.

Accuracy

$\Sigma\, |Y(i) \cap \hat{Y}(i)| \,/\, \Sigma\, |Y(i) \cup \hat{Y}(i)|$

Correct labels over all labels involved.

Precision

$\Sigma\, |Y(i) \cap \hat{Y}(i)| \,/\, \Sigma\, |\hat{Y}(i)|$

Correct labels over predicted labels.

Recall

$\Sigma\, |Y(i) \cap \hat{Y}(i)| \,/\, \Sigma\, |Y(i)|$

Correct labels over actual labels.

Hamming loss

$(1/nc)\, \Sigma\, \Sigma\, (\, I\{y \in \hat{Y}(i)\}\, I\{y \notin Y(i)\} + I\{y \notin \hat{Y}(i)\}\, I\{y \in Y(i)\}\,)$

The average number of wrong yes/no decisions over all labels.

WORKED EXAMPLE · ONE IMAGE, N = 1

Scoring one satellite image

	clear	cloudy	foggy	road	river
True Y	1	0	0	1	1
Predicted Ŷ	1	0	0	1	0

Only “river” is wrong (orange): the model missed it.

Exact match
0
sets differ

Accuracy
2/3
2 correct of 3 in the union

Precision
2/2 = 1
both predictions right

Recall
2/3
found 2 of 3 true labels

Hamming loss
1/5
1 wrong of 5 labels

SUMMARY

Five ideas to keep

1. With zero-one loss, the best classifier picks the most probable class given x.
2. Even that classifier errs where classes overlap: the Bayes error rate.
3. A confusion matrix gives tp, fp, fn and tn for every class.
4. Accuracy can hide costly mistakes: use a loss matrix, or precision, recall, specificity and Fβ.
5. Labels can overlap (multilabel) or form a tree (hierarchical), with metrics to match.

EXIT TICKET · 3 MINUTES

Check your understanding

1. Why can even the optimal classifier not reach zero error?

2. A spam filter must never send real email to the junk folder. Which metric for the spam class should you watch: precision or recall? precision bcuz it needs do know is it correct or no?

3. One image has true labels {a, b} and predicted labels {b, c}. What are its precision and recall?